

Introduction

In Part 1 of this experiment, the dataset was constrained to a narrow scope due to human limitations and restricted camera angles. The scope was reduced to four keypoints, and the dataset annotations were noisy due to human error. As a result, it was difficult to determine whether low generalization and poor prediction performance stemmed from dataset limitations or model architecture limitations.

Part 2 continues this investigation by addressing these issues through the creation of a controlled synthetic dataset in Blender.

Dataset and Setup

A Mixamo rig performing a barbell back squat (70 frames long) was imported into Blender. A camera was positioned at an angle such that the entire rig was visible, including the left and front sides of the body.

A square occluder was placed on the rig's left shoulder. It began at a scale of 0 (invisible) and increased by 0.1 scale units in both the x and y directions with each iteration. Initially, it covered only the shoulder, then gradually expanded to cover nearly the entire body except for the left wrist and left elbow.

Because the Mixamo rig is in a barbell back squat position and the occluder is placed directly on the shoulder, expanding in the same direction the model faces, the arm was not affected by occlusion. This was intentional, as it replicates Part 1, which is also based on barbell back squats.

By iteration 15, the body was fully covered and frames stopped being recorded. The dataset contains 15 occlusion levels $\times$ 70 frames, for a total of 1050 images.

A resolution of 640×480 was used.

Mean error graphs were capped at a value of 150.

Iterations and frames are 0-indexed.

Rendering and Projection

Two-dimensional keypoints were obtained by projecting 3D keypoints into image space using Blender's perspective camera model via `world_to_camera_view`.

Model

HRNet-W18 was used in Part 1 due to its feasibility for fine-tuning. HRNet-W32 was chosen for Part 2 because of its modern architecture, strong performance in pose estimation tasks, and the availability of pre-trained weights. The model was pretrained on COCO and was not fine-tuned for this experiment. It was used purely for evaluation.

Visibility Determination

A keypoint was determined to be visible using Blender’s camera ray casting. If the ray intersected the keypoint, it was considered visible.

Keypoint Definition and Ground Truth Extraction

“Left-side keypoints” were defined as the left shoulder, left elbow, left wrist, left hip, left knee, and left ankle. The dataset follows the standard COCO-17 keypoint format.

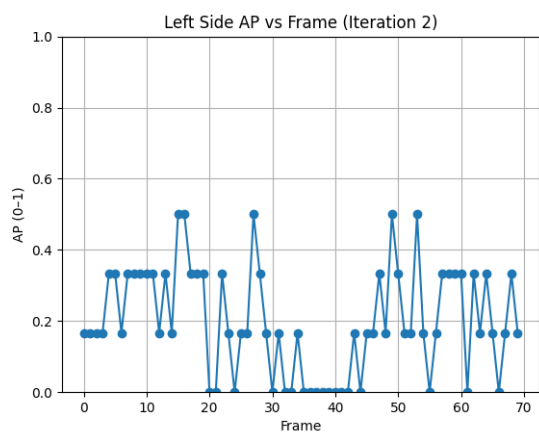
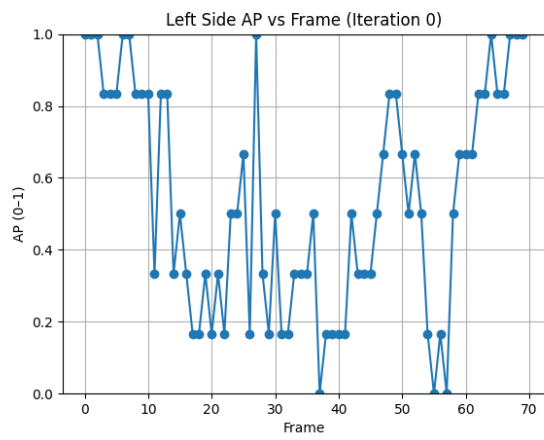
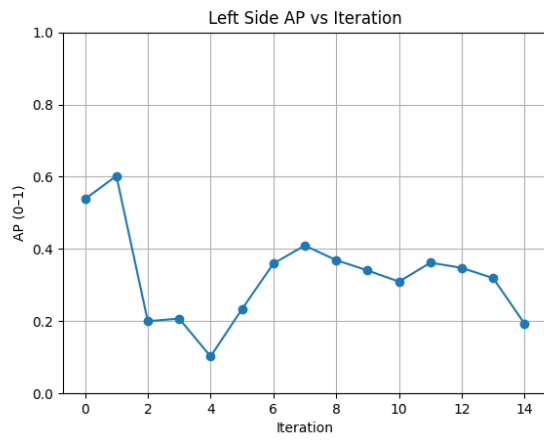
Accurate keypoints were gathered in Blender by adding icospheres to the rig at each external joint and attaching them to the model so they moved consistently during animation. While a small amount of human placement error may exist in this process, placement was consistent throughout the movement.

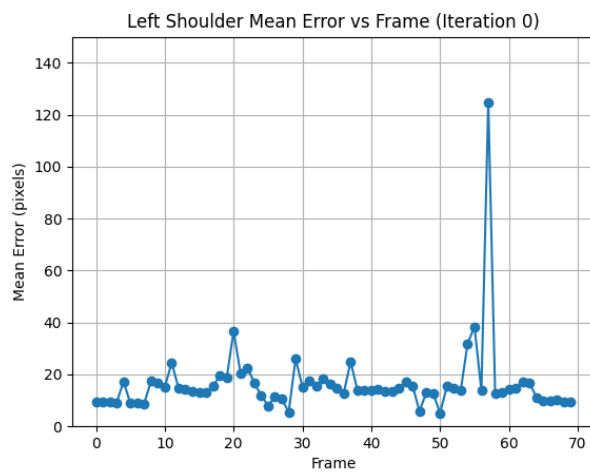
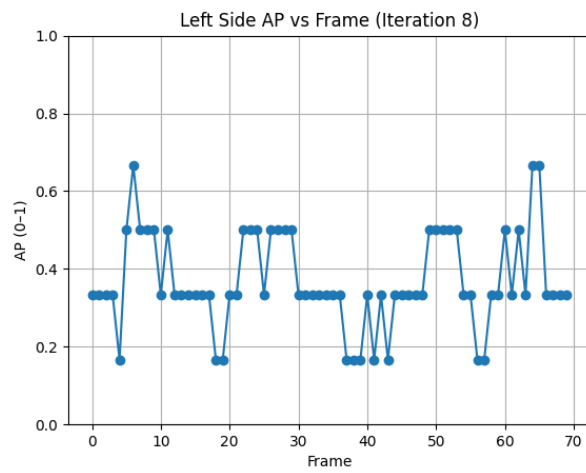
Evaluation Metrics

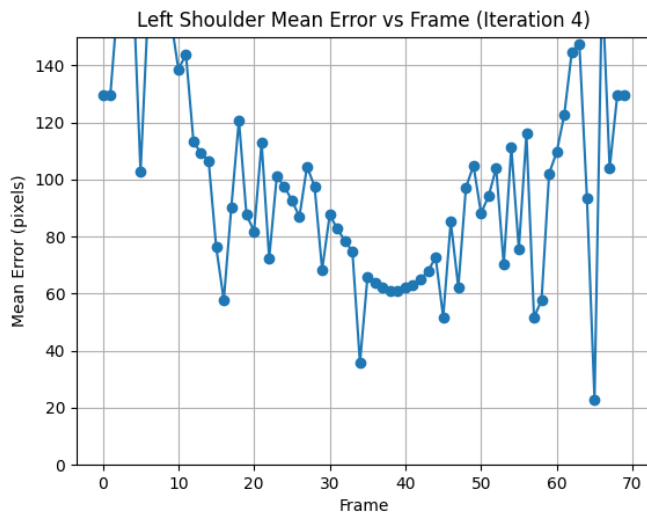
Average Precision (AP), as defined in this project, is the percentage of correctly predicted keypoints within 20 pixels of the ground truth. This differs from standard COCO mAP and instead serves as a consistent distance-based accuracy metric. It is used for all multi-keypoint graphs in the project.

For the left shoulder, per-keypoint localization error was computed using Euclidean distance in pixel space instead of AP to more accurately capture localization error for a single keypoint.

Quantitative Analysis







Qualitative Analysis

Right-side keypoints, as well as the nose, eyes, and ears, were consistently inaccurate. This was expected, as the Mixamo rig's facial structure does not resemble a normal human face, and the right side of the body is not visible to the camera.

Because of this, most graphs focus exclusively on left-side AP and exclude facial keypoints. In the first three graphs, there is a clear curve when comparing predictions of the squatted model to the standing model. AP is higher at the beginning of the movement and decreases across frames. By the fourth graph, this difference disappears, and AP remains low relative to that of iterations 1 and 2 throughout the entire movement.

Both overall AP and left-side-only AP decrease rapidly after iteration 2. Interestingly, AP increases from iteration 4 to iteration 6 and then remains approximately constant until the final iteration.

The left shoulder is evaluated using mean error distance rather than AP to better capture single-keypoint performance. Error is relatively low for the first two iterations, with occasional spikes. By iteration 3, the average error begins to increase, and by iteration 4, predictions become unreliable.

Project Scope and Limitations

- Single motion
- Single object

- Single viewpoint
 - No fine-tuning
 - Synthetic-to-real domain gap
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Results and Discussion

While Blender appears to be an ideal solution for reducing annotation noise, switching to a synthetic dataset and humanoid rig reveals another significant challenge: domain transferability. Despite facing no occlusion or very light occlusion, HRNet-W32 still struggles to produce accurate predictions. These failures are particularly pronounced when the model is in a squatted pose rather than upright.

I hypothesize that the performance stabilization observed at higher occlusion levels occurs because the network defaults to learned body priors rather than relying on image evidence.

This project suggests that domain and pose have an even greater impact on modern pose estimation performance than occlusion.